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United States Patent	12410614
Kind Code	B2
Date of Patent	September 09, 2025
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Pre-stressed insulated concrete panels and methods for making and using the same

Abstract

Pre-stressed insulated concrete panels are disclosed. Methods of making and using pre-stressed insulated concrete panels are also disclosed.

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Family ID: 83887154

Appl. No.: 18/675287

Filed: May 28, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240318426 A1	Sep. 26, 2024

Related U.S. Application Data

division parent-doc US 17723668 20220419 US 12012756 child-doc US 18675287
us-provisional-application US 63184949 20210506

Publication Classification

Int. Cl.: E04C2/28 (20060101); B28B23/04 (20060101); E04B1/14 (20060101); E04B1/16 (20060101); E04C2/288 (20060101); E04C5/08 (20060101)

U.S. Cl.:

CPC **E04C2/2885** (20130101); **B28B23/04** (20130101); **E04B1/14** (20130101); **E04B1/16** (20130101); **E04C5/08** (20130101);

Field of Classification Search

CPC: E04C (2/2885); E04C (5/08); E04C (2/06); B28B (23/04); B28B (19/003); B28B (23/0056); B28B (23/0068); B28B (23/026); B28B (23/06); E04B (1/14); E04B (1/16); E04B (1/06); E04G (21/12)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This patent application is a divisional patent application of U.S. patent application Ser. No. 17/723,668 filed on Apr. 19, 2022, and entitled “PRE-STRESSED INSULATED CONCRETE PANELS AND METHODS OF MAKING AND USING THE SAME,” now allowed, which claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 63/184,949 filed on May 6, 2021, and entitled “PRE-STRESSED INSULATED CONCRETE PANELS AND METHODS OF MAKING AND USING THE SAME,” the subject matter of both of which is hereby incorporated by reference in their entirety.

FIELD OF THE INVENTION

(1) The present invention relates generally to pre-stressed insulated concrete panels. The present invention further relates to methods of making pre-stressed insulated concrete panels and using pre-stressed insulated concrete panels such as in a building structure.

BACKGROUND OF THE INVENTION

(2) Known methods of forming pre-stressed insulated concrete panels have one or more shortcomings including, but not limited to, forming the concrete panel in a flat configuration, namely, the thickness of the concrete panel extends perpendicularly from a ground (or forming) surface, putting an enormous amount of stress on the formed concrete panel when lifting the formed concrete panel from the flat configuration, the need for a larger work area when forming formed concrete panels in the flat configuration, etc.

(3) There is a need for methods of making pre-stressed insulated concrete panels so as to minimize stress on the pre-stressed insulated concrete panels.

SUMMARY OF THE INVENTION

(4) The present invention is directed to methods of making pre-stressed insulated concrete panels so as to minimize stress on the pre-stressed insulated concrete panels. In some embodiments, the

methods of making pre-stressed insulated concrete panels comprise forming the pre-stressed insulated concrete panel with a concrete panel thickness of the pre-stressed insulated concrete panel extending in a horizontal orientation (i.e., the “thickness” of the pre-stressed insulated concrete panel being the thickness of the pre-stressed insulated concrete panel once erected at a building/structure site). The methods of making pre-stressed insulated concrete panels may further comprise forming the pre-stressed insulated concrete panels with a concrete panel width of the pre-stressed insulated concrete panel extending in a vertical orientation (i.e., the “width” of the pre-stressed insulated concrete panel being the width of the pre-stressed insulated concrete panel once erected at a building/structure site). The methods of making pre-stressed insulated concrete panels may even further comprise forming the pre-stressed insulated concrete panels with a concrete panel length of the pre-stressed insulated concrete panel extending in a horizontal orientation (i.e., the “length” of the pre-stressed insulated concrete panel being the height of the pre-stressed insulated concrete panel once erected at a building/structure site).

(5) The present invention is further directed to pre-stressed insulated concrete panels formed using the herein-disclosed methods of making pre-stressed insulated concrete panels. In some embodiments of the present invention, the pre-stressed insulated concrete panels formed using the herein-disclosed methods comprises pre-stressed insulated concrete panels comprise a concrete slab having a concrete panel width, a concrete panel length, and a concrete panel thickness, the concrete panel thickness of the concrete slab being in a horizontal orientation during formation.

(6) The pre-stressed insulated concrete panels of the present invention may further comprise (a) one or more of the following embedded panel components: (1) an insulation component, (2) (i) vertical rebar pieces, (ii) horizontal rebar pieces, or (iii) vertical rebar pieces and horizontal rebar pieces arranged in at least one layer of rebar, (3) at least one layer of fence material, (4) from about 3 to about 8 chair supports along one side surface of said concrete slab, and (5) from about 6 to about 16 pieces of pipe material, and (6) two or more sets of pre-stressed cables; (b) one or more outer surface features comprising one or more of (i) undulations and/or designs resulting from mirror undulations and/or designs along a form inner surface, (ii) one or more holes extending thru the concrete slab proximate side edges of the concrete slab, (iii) rebar pieces extending out from the outer edge surfaces of the concrete slab, (iv) one or more corner L-angle members, or (v) any combination thereof; or (c) any combination of (1) one or more embedded panel components (a) and (2) one or more outer surface features (b).

(7) The present invention is even further directed to forms for forming pre-stressed insulated concrete panels. In some embodiments of the present invention, the forms comprise (i) an upper surface of a base form, such as in the form of a beam (e.g., an I-beam), (ii) front and back side panels extending along a concrete panel width and a concrete panel length, and (iii) first and second form end panels extending along the concrete panel width and the concrete panel thickness.

(8) The present invention is also directed to kits, wherein the kits comprise the form for forming pre-stressed insulated concrete panels (e.g., the base form, and the vertically extending form components), and one or more of: a cable tensioner; a form vibrator; and a tensioning spacer.

(9) The present invention is even further directed to methods of using pre-stressed insulated concrete panels. In some embodiments of the present invention, the method of using pre-stressed insulated concrete panels comprises incorporating one or more of the pre-stressed insulated concrete panels into a wall structure. Typically, a plurality of the pre-stressed insulated concrete panels is incorporated into a given wall structure. The wall structure may comprise, for example, one or more walls of a building structure, or a continuous wall structure of a building structure, the continuous wall structure extending along an outer periphery of the building structure. In some desired embodiments, the method of using pre-stressed insulated concrete panels comprises incorporating one or more of the pre-stressed insulated concrete panels into a wall structure of a building structure comprising a residential building (e.g., an apartment structure, a single-family home, etc.), a garage, a barn, a shed, a utility building, etc.

(10) The present invention is even further directed to wall structures comprising one or more pre-stressed insulated concrete panels of the present invention. In some embodiments of the present invention, the wall structure comprises a wall structure of a building such as a residential building. (11) These and other features and advantages of the present invention will become apparent after a review of the following detailed description of the disclosed embodiments and the appended claims.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The present invention is further described with reference to the appended figure, wherein:
- (2) FIG. 1 depicts exemplary pre-stressed insulated concrete panels of the present invention;
- (3) FIG. 2 depicts concrete panel form components for forming the exemplary pre-stressed insulated concrete panels shown in FIG. 1;
- (4) FIGS. 3A-3C depict views of exemplary panel components for embedding within the exemplary pre-stressed insulated concrete panels shown in FIG. 1 (concrete is shown translucent), namely, FIG. 3A showing a side view of the exemplary panel components, FIG. 3B showing a top view of the exemplary panel components shown in FIG. 3A, and FIG. 3C showing an end view of the exemplary panel components shown in FIG. 3A;
- (5) FIG. 4 depicts a frontal view of (i) the exemplary panel components shown in FIGS. 3A-3B as if positioned on an I-beam, along a rear side panel, and between opposite end panels of the exemplary concrete panel form components shown in FIG. 2, with (ii) pre-tensioned cables extending along opposite sides of the exemplary panel components and through the various exemplary concrete panel form components (concrete is shown translucent);
- (6) FIG. 5 depicts an exemplary assembled concrete form with all internal components shown in FIG. 4 prior to pouring concrete into the exemplary assembled concrete form;
- (7) FIG. 6 depicts exemplary pre-stressed insulated concrete panels of the present invention positioned along an outer boundary ledge of a concrete slab of a building structure with temporary bracing; and
- (8) FIG. 7 depicts an aerial view of (i) numerous exemplary pre-stressed insulated concrete panels of the present invention within a staging area of a building construction site, and (ii) numerous exemplary pre-stressed insulated concrete panels of the present invention positioned along an outer boundary ledge of a concrete slab of a building structure with temporary bracing.

DETAILED DESCRIPTION OF THE INVENTION

- (9) The present invention is directed towards methods of making pre-stressed insulated concrete panels **10**. The present invention is further directed towards pre-stressed insulated concrete panels **10** obtained using the herein-described methods of making pre-stressed insulated concrete panels **10**, and methods of using the pre-stressed insulated concrete panels **10**.
- (10) I. Methods of Making Pre-Stressed Insulated Concrete Panels
- (11) The present invention is directed towards methods of making pre-stressed insulated concrete panels **10**. The disclosed methods of making pre-stressed insulated concrete panels **10** may (1) provide one or more of the following advantages over known methods of making concrete panels, and/or (2) comprise one or more of the following features: the ability to pour pre-stressed concrete panels **10** at the job site **200** using transportable concrete panel forms (e.g., forms comprising base form **20** (e.g., I-beam **20**), front and back side panels **21/22**, and first and second from end panels **23/24**); pouring pre-stressed panels **10** on site **200** utilizes concrete **60** mixed locally, thereby eliminating the need to pay expensive shipping costs to transport the heavy concrete panels poured at a remote facility; forms are designed so that the concrete panel **10** is poured on edge with the width of the panel P.sub.W oriented vertically and the length (height) of the panel P.sub.L oriented

horizontally. Since the panel **10** is taller than it is wide, the depth of concrete **60** to fill the form is reduced, as is the weight of concrete **60** that the form has to contain without bowing at the mid-span; the forms are designed on edge to ensure both the inside and outside faces of the concrete panel **10** are flat and finished (and optionally have an unlimited number of patterns and pattern options thereon), and the top and bottom faces are flat and finished and the sides and ends of the concrete panel **10** are square; the form comprises (or consists of) a base form **20** onto which are fastened form side panels **21/22**, form end panels **23/24**, and end structural beams **52/53** that anchor the pre-stressing cables **50**. The form side panels **21/22** are fastened to the form end panels **23/24**. The end structural beams **52/53** are bolted at each end of the base form **20** leaving a gap between the form end panels **23/24** and the end structural beams **52/53**. The form side panels **21/22** have horizontal and vertical structural members (i.e., grid **29**). The horizontal structural members extend beyond the form end panels **23/24** to contact the end structural beams **52/53** to support the compressive force of the pre-stressing cables **50** and the weight of the concrete **60** without allowing the side panels **21/22** to bow outward. Connecting straps are fastened between the side form panels **21/22** along the top edge to provide additional support to prevent the side panels **21/22** from bowing along the open top side of the form. After the concrete **60** is cured, the compressive force in the pre-stressed cables **50** is transferred to the concrete panel **10** by cutting the cables **50** in the gap between the form end panels **23/24** and end structural beams **52/53**; the form includes features in the base form **20** and top connecting straps to locate rebar **32/33** set within the panel **10** and allow the rebar **32/33** to extend beyond the top and bottom edge **66/67** of the concrete panel **10** (e.g., thru rebar receiving holes **35** in base form **20**); the form end panels **23/24** and end structural beams **52/53** include holes **54/55** through which the pre-stressing cables **50** are passed; reusable pre-stressing cable anchors **51** are installed abut to the outside face of the end support beams **52/53** to hold the tension in the cables **50** after tensioning; form end panels **23/24** include rare earth magnets (not shown) at the top and bottom to which steel L-angle clips **80** can be attached; forms include integral vibrators (not shown) to ensure concrete **60** flows to fill all the open space thus avoiding voids; patterns (not shown) may be incorporated onto the inside face of the forms to simulate brick, stone, wood boards, etc. to customize the exterior or both faces of the wall **100**; multiple concrete panel forms may be included on the same base beam **20** as long as a gap is included between the form end panel **24** of one form and the form end panel **23** of the adjacent concrete panel form to allow the pre-stressing cables **50** to be cut after the concrete panels **10** cure; after the concrete **60** is cured and the pre-stressing cables **50** are cut, the form side panels **21/22** and end panels **23/24** are unbolted and removed to strip the concrete panel **60** from the form; after the pre-stressed concrete panel **10** is removed from the form, the form can be reassembled and the internal elements reloaded and poured again. Concrete panels **10** can be poured in batches to minimize the total number of transportable forms required to pour all the needed concrete panels **10**; the ability to move/transport a concrete panel **10**, for example, by using only the internal rebar **32** of the concrete panel **10** or by clamping onto the concrete panel **10** (i.e., so as to eliminate the need for additional rigging eyes that are embedded into the face of the panel and on the top side of traditionally made, flat concrete panels). The worst loading condition for panels that are poured flat (e.g., traditional pre-stressed concrete panels) is lifting them from the flat position to a vertical position. This requires additional rigging eyes be embedded into the face of the panel and on the top side of the panel. Lifting the top edge while resting on the bottom edge puts the length of the panel under bending stress due to its own weight. The section modulus of the panel is weakest in this direction and most cracks occur when up righting the panel; this bending problem (with traditional panels) is eliminated by lifting the panel **10** on its side. Specially designed lifting fixtures are fastened to the rebar **32** extending from the top surface **69** (side of the panel **10**) of the panel **10** for stripping it from the form; the panel **10** is then transported to a storage rack **75** to allow the panel **10** to finish curing while on its side; after the panel **10** is fully cured, the panel **10** is lifted and placed in a rotary device (not shown) that rotates it 90° to its final upright position; lifting fixtures (not shown) are mounted to

the rebar **32** extending from both sides **66/67** of the panel **10** for transport to its final position on the foundation panel ledge **30**; and the addition of lifting eyes embedded into the face and top of the panel is eliminated. This eliminates the need to fill in the concrete on the face of the panels over these embedded lifting eyes.

II. Pre-Stressed Insulated Concrete Panels

(12) The present invention is also directed to pre-stressed insulated concrete panels **10** formed via the herein-described methods of forming pre-stressed insulated concrete panels **10**. The pre-stressed insulated concrete panels **10** include, but are not limited to, the pre-stressed insulated concrete panels **10** described in the specification, the numerous embodiments, and the claims below.

(13) The disclosed pre-stressed insulated concrete panels **10** may comprise one or more of the following features: insulation board **31'** centered between the wythes of concrete **60** that forms the inside and outside faces of the panel **10**. Insulation **31** is centered between the left and right sides and top and bottom of the concrete panel **10**; solid concrete **60** on the top and bottom and right and left sides of the concrete panel **10**; rebar **32** that extends from both sides **66/67** of the panel **10** is located on either side of the insulation **31**. Rebar **32** is used to lift and transport the concrete panels **10** and provide anchorage for additional rebar added in the gaps between adjacent concrete panels **10** as described herein; pre-stressing cables **50** that extend from the top to the bottom of the concrete panel **10** and located on either side of the insulation **31**; fence material **36** that extends the width and length (height) of the concrete panel **10** is located on either side of the insulation **31**; pins **41** that extend thru the insulation **31** in a grid pattern that include a feature to anchor the inside and outside concrete wythes **82** to the insulation **31**; holes **65** along and proximate the left and right-side edges **66/67** to be used for bolting the interior and exterior pour-in-place forms (not shown) (e.g., that straddle gaps between panels **10**) tightly against the inside and outside faces of the concrete panels **10**; steel L-angle clips **80** at the bottom edge corners of the panel **10** to be used to weld/anchor the concrete panels **10** to the slab **30**; pre-stressed cables **50** provide compressive stress in concrete panels **10**; pre-stressed cables **50** run from top to bottom of the concrete panel **10** to provide compressive stress to concrete panel **10** to increase bending strength and greater resistance to wind and seismic loads; pre-stressed cables **50** are installed on either side of the insulation **31**; pre-stressed cables **50** are stressed in pairs to eliminate side load bending on the form and within the concrete panel; and form only has to resolve compressive load due to stressing the cables **50**.

III. Methods of Using Pre-Stressed Insulated Concrete Panels

(14) The present invention is also directed towards methods of using pre-stressed insulated concrete panels **10**. Methods of using pre-stressed insulated concrete panels **10** include methods of using the pre-stressed insulated concrete panels **10** formed via the herein-described methods of making pre-stressed insulated concrete panels **10**. The methods of using pre-stressed insulated concrete panels **10** may include, but are not limited to, methods of forming a building structure **200**, a method of forming a wall structure **100**, etc.

(15) In one exemplary embodiment, the method of using the pre-stressed insulated concrete panels **10** comprises forming a building structure **200**, wherein at least one wall **100** of the building structure **200** comprises a pre-stressed insulated concrete panel **10** of the present invention. Suitable building structures **200** include, but are not limited to, a residential building (e.g., an apartment structure, a single-family home, etc.), a garage, a barn, a shed, a utility building, an electrical/computer housing building, a commercial building, an industrial building, any other structure that is desired to be insulated, etc.

(16) The present invention will be further described in the following additional embodiments, examples, and claims.

Additional Embodiments

(17) Methods of Making Pre-Stressed Insulated Concrete Panels

(18) 1. A method of making a pre-stressed insulated concrete panel **10**, said method comprising: forming the pre-stressed insulated concrete panel **10** with a concrete panel thickness $P_{sub.T}$ of the pre-stressed insulated concrete panel **10** extending in a horizontal orientation (i.e., a horizontal orientation relative to a ground surface **30** or a base surface **30** or a concrete slab, etc.). As discussed above, the concrete panel thickness $P_{sub.T}$ of a given pre-stressed insulated concrete panel **10** refers to the “thickness” of the pre-stressed insulated concrete panel **10** once erected at a building/structure site. See, for example, the “thickness” $P_{sub.T}$ of the pre-stressed insulated concrete panel **10** shown in FIG. 1.

(19) 2. The method of embodiment 1, wherein said forming step comprises: forming the pre-stressed insulated concrete panel **10** with a concrete panel width $P_{sub.W}$ of the pre-stressed insulated concrete panel **10** extending in a vertical orientation (i.e., vertical orientation relative to a ground surface **30** or a base surface **30** or a concrete slab, etc.). As discussed above, the concrete panel width $P_{sub.W}$ of a given pre-stressed insulated concrete panel **10** refers to the “width” of the pre-stressed insulated concrete panel **10** once erected at a building/structure site. See, for example, the width $P_{sub.W}$ of the pre-stressed insulated concrete panel **10** shown in FIG. 1.

(20) 3. The method of embodiment 1 or 2, wherein said forming step comprises: forming the pre-stressed insulated concrete panel **10** with a concrete panel length (height) $P_{sub.L}$ of the pre-stressed insulated concrete panel **10** extending in the horizontal orientation (i.e., relative to a ground surface **30** or a base surface **30** or a concrete slab, etc.). As discussed above, the concrete panel length $P_{sub.L}$ of a given pre-stressed insulated concrete panel **10** refers to the “height” of the pre-stressed insulated concrete panel **10** once erected at a building/structure site. See, for example, the length $P_{sub.L}$ of the pre-stressed insulated concrete panel **10** shown in FIG. 1.

(21) 4. The method of any one of embodiments 1 to 3, wherein said forming step comprises: forming the pre-stressed insulated concrete panel **10** on a base form **20** (e.g., a beam **20** such as an I-beam **20**). It should be understood that other base forms **20** and/or forming members/substrates having an upper surface similar to upper surface **26** of base form **20** (e.g., I-beam **20**) could be used in place of base form **20** (e.g., I-beam **20**).

(22) 5. The method of embodiment 4, wherein the base form **20** (e.g., I-beam **20**) extends horizontally, substantially parallel with a ground surface **30**.

(23) 6. The method of embodiment 4 or 5, wherein said forming step comprises: forming the pre-stressed insulated concrete panel **10** on (i) a base form **20** (e.g., I-beam **20**) and between (ii) front and back side panels **21/22** extending along the concrete panel width $P_{sub.W}$ and the concrete panel length (height) $P_{sub.L}$, and (ii) first and second form end panels **23/24** extending along the concrete panel width $P_{sub.W}$ and the concrete panel thickness $P_{sub.T}$.

(24) 7. The method of embodiment 6, wherein the base form **20** (e.g., I-beam **20**) comprises one or more first base form holes **25** within an upper surface **26** of the base form **20** (e.g., I-beam **20**), and said forming step comprises attaching the first and second form end panels **23/24** to the upper surface **26** of the base form **20** (e.g., I-beam **20**) via the one or more first base form holes **25**.

(25) 8. The method of embodiment 6 or 7, wherein the base form **20** (e.g., I-beam **20**) comprises one or more second base form holes **27** within the upper surface **26** of the base form **20** (e.g., I-beam **20**), and said forming step comprises attaching the first and second side panels **21/22** to the base form **20** (e.g., I-beam **20**) via the one or more second base form holes **27**. See, for example, the exemplary form components shown in FIG. 2.

(26) 9. The method of any one of embodiments 6 to 8, wherein each of the front and back side panels **21/22** comprises an aluminum sheet **28**. It should be understood that front and back side panels **21/22** may comprise materials other than aluminum. Suitable other materials include, but are not limited to, other metals, fiberglass, polymers, fiber-reinforced polymers, ceramics, thermosettable panels, etc.

(27) 10. The method of embodiment 9, wherein each aluminum sheet **28** is backed with a metal grid **29**.

- (28) 11. The method of any one of embodiments 6 to 10, wherein each of the front and back side panels **21/22** has a side panel height SP.sub.H (i.e., extending in a vertical direction) of from about 30 inches (in) to about 72 in.
- (29) 12. The method of any one of embodiments 6 to 11, wherein each of the front and back side panels **21/22** has a side panel height SP.sub.H (i.e., extending in a vertical direction V) of from about 36 in to about 62 in.
- (30) 13. The method of any one of embodiments 6 to 12, wherein each of the front and back side panels **21/22** has a side panel length SP.sub.L (i.e., extending in a horizontal direction H) of from about 8 ft to about 12 ft.
- (31) 14. The method of any one of embodiments 6 to 13, wherein each of the front and back side panels **21/22** has a side panel length SP.sub.L (i.e., extending in a horizontal direction H) of from about 9.5 ft to about 10 ft.
- (32) 15. The method of any one of embodiments 6 to 14, wherein each of the first and second form end panels **23/24** comprises an aluminum post **23/24**. It should be understood that first and second form end panels **23/24** may comprise materials other than aluminum. Suitable other materials include, but are not limited to, other metals, fiberglass, polymers, fiber-reinforced polymers, ceramics, thermosettable panels, etc.
- (33) 16. The method of any one of embodiments 6 to 15, wherein each of the first and second form end panels **23/24** has an end panel height EP.sub.H (i.e., extending in a vertical direction V) of from about 30 in to about 72 in.
- (34) 17. The method of any one of embodiments 6 to 16, wherein each of the first and second form end panels **23/24** has an end panel height EP.sub.H (i.e., extending in a vertical direction V) of from about 36 in to about 62 in.
- (35) 18. The method of any one of embodiments 6 to 17, wherein each of the first and second form end panels **23/24** has an end panel width EP.sub.W (i.e., extending in a horizontal direction H) of from about 6 in to about 16 in.
- (36) 19. The method of any one of embodiments 6 to 18, wherein each of the first and second form end panels **23/24** has an end panel width EP.sub.W (i.e., extending in a horizontal direction H) of from about 6 in to about 12 in. In some embodiments, end panel width EP.sub.W is about 7.0 in.
- (37) 20. The method of any one of embodiments 4 to 19, wherein the base form **20** (e.g., I-beam **20**) has a base form length IB.sub.L of from about 10 feet (ft) to about 24 ft.
- (38) 21. The method of any one of embodiments 4 to 20, wherein the base form **20** (e.g., I-beam **20**) has a base form length IB.sub.L of about 24 ft.
- (39) 22. The method of any one of embodiments 4 to 21, wherein the base form **20** (e.g., I-beam **20**) has a base form width IB.sub.W of from about 10.0 in to about 24 in. In some embodiments, base form width IB.sub.W is about 12.0 in.
- (40) 23. The method of any one of embodiments 6 to 22, wherein said forming step further comprises: positioning one or more embedded panel components (discussed below and shown in FIGS. 3A-5) within a form volume F.sub.V bound by (i) an upper surface **26** of the base form **20** (e.g., I-beam **20**), (ii) inner surfaces **211/221** of the front and back side panels **21/22**, and (iii) inner surfaces **231/241** of the first and second form end panels **23/24**.
- (41) 24. The method of embodiment 23, wherein the one or more embedded panel components comprise an insulation component **31**.
- (42) 25. The method of embodiment 24, wherein the insulation component **31** comprises an insulation board **31'**.
- (43) 26. The method of any one of embodiments 23 to 25, wherein the one or more embedded panel components comprise (i) vertical rebar pieces **32**, (ii) horizontal rebar pieces **33**, or (iii) vertical rebar pieces **32** and horizontal rebar pieces **33** arranged in at least one layer of rebar **32/33**.
- (44) 27. The method of embodiment 26, wherein the embedded panel components comprise vertical rebar pieces **32**, and said positioning step further comprises inserting end portions **321** of

the vertical rebar pieces **32** thru receiving holes **35** on an upper surface **26** of the base form **20** (e.g., I-beam **20**).

(45) **28**. The method of embodiment **27**, wherein the one or more embedded panel components comprise horizontal rebar pieces **33**, and said positioning step further comprises attaching the horizontal rebar pieces **33** to the vertical rebar pieces **32**.

(46) **29**. The method of any one of embodiments **26** to **28**, wherein the at least one layer of rebar **32/33** comprises two layers of rebar **32/33** along opposite sides **311/312** of the insulation component **31**.

(47) **30**. The method of any one of embodiments **23** to **29**, wherein the one or more embedded panel components comprise at least one layer of fence material **36**.

(48) **31**. The method of embodiment **30**, wherein said positioning step further comprises attaching the at least one layer of fence material **36** to (i) the vertical rebar pieces **32**, (ii) the horizontal rebar pieces **33**, or (iii) the vertical rebar pieces **32** and the horizontal rebar pieces **33**.

(49) **32**. The method of embodiment **30** or **31**, wherein at least one layer of fence material **36** comprises two layers of fence material **36/36** along opposite sides **311/312** on the insulation component **31**.

(50) **33**. The method of embodiment **32**, wherein the insulation component **31** comprises the insulation board **31'** which is sandwiched between the two layers of rebar **32/33**, and the two layers of rebar **32/33** are sandwiched between the two layers of fence material **36/36**.

(51) **34**. The method of any one of embodiments **23** to **33**, wherein the one or more embedded panel components comprise from about 3 to about 8 chair supports **37**, and said positioning step further comprises positioning the chair supports **37** along an upper surface **26** of the base form **20** (e.g., I-beam **20**), the chair supports **37** providing support for the insulation component **31** (e.g., insulation board **31'**) while concrete is being poured onto the insulation component **31**.

(52) **35**. The method of any one of embodiments **23** to **34**, wherein the one or more embedded panel components comprise from about 6 to about 16 pieces of pipe material **38**, and said positioning step further comprises positioning the pieces of pipe material **38** thru holes **40** along top and bottom edges **222/223** of the front and back side panels **21/22**.

(53) **36**. The method of any one of embodiments **23** to **35**, wherein the one or more embedded panel components comprise a plurality of wythe tie pins **41**, and said positioning step further comprises installing the plurality of wythe tie pins **41** in a grid pattern through pre-drilled holes **42** extending through the insulation board **31'**.

(54) **37**. The method of any one of embodiments **1** to **35**, wherein said forming step further comprises: positioning a plurality of cables **50** so as to extend thru (a) a first pre-stressing cable anchor **51**, a first end support beam **52**, the first form end panel **23**, the second form end panel **24**, and out thru an opposite second end support beam **53**.

(55) **38**. The method of embodiment **37**, wherein said positioning step further comprises looping the plurality of cables **50** back thru the opposite second end support beam **43**, the second form end panel **24**, the first form end panel **23**, the first end support beam **52**, and a second pre-stressing cable anchor **51**. See, for example, FIGS. **4-5**.

(56) **39**. The method of embodiment **37** or **38**, further comprising: applying tension on the plurality of cables **50**.

(57) **40**. The method of embodiment **39**, wherein said applying step comprises applying tension on a loop portion **50'** of each cable **50** within the plurality of cables **50**, either simultaneously or sequentially. Note that this is another, important differentiator from known methods of tensioning cables wherein the cables are tensioned individually.

(58) **41**. The method of any one of embodiments **1** to **40**, wherein said forming step further comprises: pouring concrete **60** into a form comprising (i) a base form **20** (e.g., I-beam **20**), (ii) front and back side panels **21/22** extending along a concrete panel width P.sub.W and a concrete panel length P.sub.L, and (iii) first and second form end panels **23/24** extending along the concrete

panel width P.sub.W and the concrete panel thickness P.sub.T.

(59) 42. The method of embodiment 41, further comprising: allowing the concrete **60** to cure.

(60) 43. The method of any one of embodiments 4 to 42, wherein said forming step comprises forming two pre-stressed insulated concrete panels **10** along the base form **20** (e.g., I-beam **20**) during a single batch formation step.

(61) 44. The method of embodiment 43, wherein said forming step comprises forming each of the two pre-stressed insulated concrete panels **10** on (i) the base form **20** (e.g., I-beam **20**) and between (ii) two sets of front and back side panels **21/22** extending along a concrete panel width P.sub.W and a concrete panel length P.sub.L, and (ii) two sets of first and second form end panels **23/24** extending along the concrete panel width P.sub.W and the concrete panel thickness P.sub.T.

(62) 45. The method of any one of embodiments 41 to 44, further comprising: separating the pre-stressed insulated concrete panel **10** from the form **20/21/22/23/24**.

(63) 46. The method of any one of embodiments 4 to 45, wherein the method takes place at a job site **70** (e.g., a single-family home construction site **70**). See, for example, exemplary job site **70** shown in FIG. 7.

(64) 47. A method of making a pre-stressed insulated concrete panel **10**, said method comprising: forming the pre-stressed insulated concrete panel **10** at a job site **70** using transportable form components, the transportable form components comprising: a base form **20** (e.g., a beam **20** such as I-beam **20**), first and second form end panels **23/24**, and first and second form end panels **23/24**.

(65) 48. The method of embodiment 47, wherein said forming step comprises forming the pre-stressed insulated concrete panel **10** with a concrete panel thickness P.sub.T of the pre-stressed insulated concrete panel **10** extending in a horizontal orientation (i.e., a horizontal orientation relative to a ground surface **30** or a base surface **30** or a concrete slab, etc.).

(66) 49. The method of embodiment 47 or 48, wherein said method comprises one or more features and/or steps as described in any one of embodiments 1 to 46.

(67) Pre-Stressed Insulated Concrete Panels

(68) 50. A pre-stressed insulated concrete panel **10** formed in the method of any one of embodiments 4 to 49.

(69) 51. A pre-stressed insulated concrete panel **10** comprising: a concrete slab **10'** having a concrete panel width P.sub.W, a concrete panel length P.sub.L, and a concrete panel thickness P.sub.T, the concrete panel thickness P.sub.T of the concrete slab **10'** being in a horizontal orientation during formation.

(70) 52. The pre-stressed insulated concrete panel **10** of embodiment 50 or 51, wherein the concrete panel width P.sub.W of the concrete slab **10'** being in a vertical orientation during formation, and the concrete panel length P.sub.L, of the concrete slab **10'** being in a horizontal orientation during formation.

(71) 53. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 52, wherein the concrete slab **10'** comprises one or more embedded panel components, said one or more embedded panel components comprising one or more of: (1) an insulation component **10**, (2) (i) vertical rebar pieces **32**, (ii) horizontal rebar pieces **33**, or (iii) vertical rebar pieces **32** and horizontal rebar pieces **33** arranged in at least one layer of rebar **32/33**, (3) at least one layer of fence material **36**, (4) from about 3 to about 8 chair supports **37** along one side surface **68** of said concrete slab **10'**, and (5) from about 6 to about 16 pieces of pipe material **38**. It should be noted that in some embodiments, chair supports **37** are permanently embedded within the concrete slab **10'**. In other embodiments, chair supports **37** are removable from the concrete slab **10'**.

(72) 54. The pre-stressed insulated concrete panel **10** of embodiment 50 or 53, wherein the one or more embedded panel components comprise an insulation component **31**.

(73) 55. The pre-stressed insulated concrete panel **10** of embodiment 54, wherein the insulation component **31** comprises an insulation board **31'**.

(74) 56. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 and 53 to 55,

wherein the one or more embedded panel components comprise at least one layer of rebar **32/33**.

(75) 57. The pre-stressed insulated concrete panel **10** of embodiment 56, wherein the at least one layer of rebar **32/33** comprises two layers of rebar **32/33** along opposite sides **311/312** on the insulation component **31**.

(76) 58. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 and 53 to 57, wherein the one or more embedded panel components comprise at least one layer of fence material **36**.

(77) 59. The pre-stressed insulated concrete panel of embodiment 58, wherein the at least one layer of fence material **36** comprises two layers of fence material **36/36** along opposite sides **311/312** of the insulation component **31**.

(78) 60. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 and 53 to 59, wherein the one or more embedded panel components comprise (1) an insulation board **31'**, (2) vertical rebar pieces **32** and horizontal rebar pieces **33** arranged in at least one layer of rebar **32/33**, and (3) at least one layer of fence material **36**, wherein the insulation board **31'** is sandwiched between two layers of rebar **32/33**, and the two layers of rebar **32/33** are sandwiched between two layers of fence material **36/36**.

(79) 61. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 and 53 to 60, wherein the one or more embedded panel components comprise from about 3 to about 8 chair supports **37**, and the chair supports **37** are along an outer side edge surface **68** of the concrete slab **10'**.

(80) 62. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 and 53 to 61, wherein the one or more embedded panel components comprise from about 6 to about 16 pieces of pipe material **38**, and the pieces of pipe material **38** form holes **65** extending along outer side edges **66/67** of the concrete slab **10'**.

(81) 63. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 and 53 to 62, wherein the one or more embedded panel components comprise a plurality of wythe tie pins **41**, the plurality of wythe tie pins **41** being in a grid pattern through pre-drilled holes **42** extending through the insulation board **31'**.

(82) 64. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 63, wherein the concrete slab **10'** comprises one or more surface features on one or more outer surfaces **108/109** thereof, said surface features comprising one or more of (i) undulations and/or designs (not shown) resulting from mirror undulations and/or designs along a form inner surface (e.g., upper surface **26** of base form **20** (e.g., I-beam **20**), or inner surface **221** of back side panel **22**), (ii) one or more holes **65** extending thru the concrete slab **10'** proximate side edges **66/67** of the concrete slab **10'**, (iii) rebar pieces **32** extending out from outer side surfaces **68/69** of the concrete slab **10'**, (iv) one or more corner L-angle members **80**, or (v) any combination thereof. As discussed above, the surface features may include a pattern or patterns on one or more of outer surfaces **108/109**. Any pattern and/or patterns may be used.

(83) 65. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 64, wherein the concrete slab **10'** comprises one or more surface features comprising undulations and/or designs (not shown) resulting from mirror undulations and/or designs along a form inner surface (e.g., upper surface **26** of base form **20** (e.g., I-beam **20**), or inner surface **221** of back side panel **22**).

(84) 66. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 65, wherein the concrete slab **10'** comprises one or more surface features comprising one or more holes **65** extending thru the concrete slab **10'** proximate side edges **66/67** of the concrete slab **10'**.

(85) 67. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 66, wherein the concrete slab comprises one or more surface features comprising rebar pieces **32** extending out from outer side surfaces **68/69** of the concrete slab **10'**.

(86) 68. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 67, wherein the concrete slab **10'** comprises one or more surface features comprising two corner L-angle

members **80** in corners along a bottom edge **81** of the concrete slab **10'**.

(87) 69. The pre-stressed insulated concrete panel **10** of any one of embodiments 64 to 67, wherein the form comprises (i) an upper surface **26** of a base form **20** (e.g., I-beam **20**), (ii) front and back side panels **21/22** extending along the concrete panel width P.sub.W and the concrete panel length (height) P.sub.L, and (iii) first and second form end panels **23/24** extending along the concrete panel width P.sub.W and the concrete panel thickness P.sub.T.

(88) 70. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 69, wherein the concrete panel width P.sub.W ranges from about 24 in to about 72 in, the concrete panel length P.sub.L ranges from about 72 in to about 130 in, and the concrete panel thickness P.sub.T ranges from about 5.0 in to about 14.0 in.

(89) 71. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 70, wherein the concrete panel width P.sub.W ranges from about 30 in to about 60 in, the concrete panel length (height) P.sub.L ranges from about 110 in to about 120 in, and the concrete panel thickness P.sub.T ranges from about 6.0 in to about 12.0 in.

(90) 72. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 71,

(91) wherein (a) the concrete panel width P.sub.W is about 36 in, the concrete panel length (height) P.sub.L is about 116 in, and the concrete panel thickness P.sub.T is about 7.0 in, or (b) the concrete panel width P.sub.W is about 60 in, the concrete panel length (height) P.sub.L is about 116 in, and the concrete panel thickness P.sub.T is about 7.0 in.

(92) 73. The pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 72, wherein the pre-stressed insulated concrete panel **10** does not have any panel lifting eyes installed on or within either of major surfaces **108/109** of the pre-stressed insulated concrete panel **10**. See, FIG. 6.

(93) Forms for Forming Using Pre-Stressed Insulated Concrete Panels

(94) 74. A form for use in the method of any one of embodiments 1 to 49 or for forming the pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 73, wherein the form comprises (i) an upper surface **26** of an base form **20** (e.g., I-beam **20**), (ii) front and back side panels **21/22** extending along the concrete panel width P.sub.W and the concrete panel length (height) P.sub.L, and (iii) first and second form end panels **23/24** extending along the concrete panel width P.sub.W and the concrete panel thickness P.sub.T.

(95) Kits for Forming Using Pre-Stressed Insulated Concrete Panels

(96) 75. A kit comprising the form of embodiment 74, said kit further comprising one or more of: a cable tensioner (not shown); a form vibrator (not shown); a tensioning spacer; (not shown), a lifting device (not shown), a rotating device (not shown), and a storage rack **75**.

(97) Methods of Using Pre-Stressed Insulated Concrete Panels

(98) 76. A method of using the pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 73, said method comprising: incorporating one or more of the pre-stressed insulated concrete panels **10** into a wall structure **100**.

(99) 77. The method of embodiment 76, wherein said incorporating step comprises: incorporating a plurality of the pre-stressed insulated concrete panels **10** into the wall structure **100**.

(100) 78. The method of embodiment 76 or 77, wherein the wall structure **100** comprises one or more walls **101** of a building structure **200**.

(101) 79. The method of any one of embodiments 76 to 78, wherein the wall structure **100** comprises a continuous wall structure **120** of a building structure **200**, the continuous wall structure **120** extending along an outer periphery **201** of the building structure **200**.

(102) 80. The method of embodiment 78 or 79, wherein the building structure **200** comprises a residential building **200** (e.g., an apartment structure, a single-family home, etc.), a garage, a barn, a shed, a utility building, an electrical/computer housing building, a commercial building, an industrial building, any other structure that is desired to be insulated, etc.

(103) 81. The method of any one of embodiments 78 to 80, further comprising: spacing the one or more pre-stressed insulated concrete panels **10** apart from one another along a base surface **30** of

the building structure **200**.

(104) 82. The method of embodiment 81, wherein the one or more pre-stressed insulated concrete panels **10** are spaced apart (i) by about 1.0 ft to 1.5 ft along wall portions **121** that do not contain a window (not shown) or door (not shown), or (ii) by about 2.0 ft to about 10.0 along wall portions **122** that do contain a window (not shown) or door (not shown).

(105) 83. The method of embodiment 81 or 82, wherein the one or more pre-stressed insulated concrete panels **10** are spaced apart (i) by about 1.0 ft along wall portions **121** that do not contain a window or door, or (ii) by about 3.0 ft to about 6.0 along wall portions **122** that do contain a window or door.

(106) 84. The method of any one of embodiments 81 to 83, further comprising: positioning one or more wall embedded components (not shown) between the spaced apart pre-stressed insulated concrete panels **10**, the one or more wall embedded components comprising electrical wiring (not shown), electrical outlets (not shown), wall insulation component (not shown), wall rebar (not shown), plumbing (not shown), chases (not shown), fiber-optic cable (not shown), heating and cooling components (not shown), or any combination thereof.

(107) 85. The method of any one of embodiments 81 to 84, further comprising: attaching pour-in-place forms (not shown) to the spaced apart pre-stressed insulated concrete panels **10**.

(108) 86. The method of embodiment 85, wherein said attaching step comprises attaching the pour-in-place forms to the spaced apart pre-stressed insulated concrete panels **10** via holes **65** along outer edges **66/67** of the pre-stressed insulated concrete panels **10**.

(109) 87. The method of embodiment 85 or 86, further comprising: pouring concrete **60** in gaps **121** between the spaced apart pre-stressed insulated concrete panels **10** and the pour-in-place forms (not shown) to form the continuous wall structure **120**.

(110) 88. The method of any one of embodiments 85 to 87, further comprising: removing the pour-in-place forms (not shown) from the wall structure **100**.

(111) 89. The method of embodiment 78 or 88, further comprising: assembling a roof (not shown) onto the one or more walls **101** of the building structure **200**.

(112) Wall Structures Comprising Pre-Stressed Insulated Concrete Panels

(113) 90. A wall structure **100** comprising the pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 73.

(114) 91. A wall structure **100** comprising a plurality of pre-stressed insulated concrete panels **10**, wherein each of the pre-stressed insulated concrete panels **10** comprises the pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 73.

(115) 92. A wall structure **100** comprising (i) a plurality of spaced apart pre-stressed insulated concrete panels **10**, wherein each of the pre-stressed insulated concrete panels **10** comprises the pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 73, and (ii) pour-in-place concrete columns (not shown) between the spaced apart pre-stressed insulated concrete panels **10**.

(116) 93. A wall structure **100** comprising (i) a plurality of spaced apart pre-stressed insulated concrete panels **10**, wherein each of the pre-stressed insulated concrete panels **10** comprises the pre-stressed insulated concrete panel **10** of any one of embodiments 50 to 73, (ii) pour-in-place concrete columns (not shown) between some of the spaced apart pre-stressed insulated concrete panels **10**, and (iii) pour-in-place headers and/or footers (not shown) between some of the spaced apart pre-stressed insulated concrete panels **10**.

(117) Building Structures Comprising Pre-Stressed Insulated Concrete Panels

(118) 94. A building **200** comprising the wall structure **100** of any one of embodiments 90 to 93.

(119) 95. The building **200** of embodiment 94, wherein the building **200** is a residential building **200** (i.e., a building occupied by people). As discussed above, the residential building **200** could be any building **200** in which people reside such as an apartment structure, a single-family home, etc.

(120) 96. The building **200** of embodiment 94 or 95, wherein the building **200** is a single-family home **200**.

(121) 97. The building **200** of embodiment 94, wherein the building **200** is a garage, a barn, a shed, a utility building, an electrical/computer housing building, a commercial building, an industrial building, or any other type of building in which people and/or animals use (but not necessarily reside in).

(122) The present invention is further illustrated by the following example, which is not to be construed in any way as imposing limitations upon the scope thereof. On the contrary, it is to be clearly understood that resort may be had to various other embodiments, modifications, and equivalents thereof which, after reading the description herein, may suggest themselves to those skilled in the art without departing from the spirit of the present invention and/or the scope of the appended claims.

EXAMPLES

(123) The following examples provide techniques for forming pre-stressed insulated concrete panels **10**, and using the pre-stressed insulated concrete panels **10** to form a building structure **200**.
Example 1—Forming Pre-Stressed Insulated Concrete Panels

(124) The following steps were used to form pre-stressed insulated concrete panels **10** at a home construction site **200**:

(125) 1. Concrete panel forms, namely, (i) a base form **20** (e.g., I-beam **20**), (ii) opposite front and back side panels (i.e., back and front side panels **21/22**), and (iii) opposite first and second form end panels **23/24**, were assembled except for the front form side panel **21** (i.e., in a manner similar to the exemplary form and exemplary panel components set-up shown in FIG. 4);

(126) 2. Internal panel wall structural components were installed within the form from the back side panel **22** to the front side panel **21**, with the internal panel wall structural components comprising (i) a first set of pre-tensioned $\frac{3}{8}$ in cables **50**, (ii) a first layer of fence material **36**, (iii) a first layer of $\frac{1}{2}$ in rebar **32/33**, (iv) a 3 in thick rigid insulation board **31'**, (v) a second grid of $\frac{1}{2}$ in rebar **32/33**, (vi) a second layer of fence material **36**, and (vii) a second set of pre-tensioned $\frac{3}{8}$ in cables **50**;

(127) 3. In step (2) above, release cones (not shown) were installed onto the end portions **321** of all vertical rebar pieces **32** that penetrated the bottom base form **20** (e.g., I-beam **20**);

(128) 4. In step (2) above, each layer of rebar **32/33** was installed so that (a) end portions **321** of the vertical rebar pieces **32** extended into (i) holes **35** within the base form **20** and (ii) holes in connecting straps, and (b) horizontal rebar pieces **33** were positioned and secured onto the vertical rebar pieces **32** (e.g., via wire (not shown)). (Note, a horizontal piece of steel flat bar with predrilled holes therein (not shown) may be placed on top of front side panel **21** and connect with a top of back side panel **22**. The horizontal piece of steel flat bar can slide down over the vertical rebar pieces **32** (i.e., the rebar ends opposite end portions **321** of the vertical rebar pieces **32**) so that the vertical rebar pieces **32** extend through the predrilled holes of the horizontal piece of steel flat bar. When fastened to front side panel **21** and back side panel **22**, the horizontal piece of steel flat bar holds the captured rebar ends **32** (i.e., the rebar ends opposite end portions **321** of the vertical rebar pieces **32**) in the correct position.);

(129) 5. In step (2) above, each fence **36** was wired to adjacent rebar layers **32/33** so as to center the fence **36**, left/right and top/bottom, within the form;

(130) 6. In step (2) above, temporary insulation support chairs **37** were installed to support the insulation board **31'**; the insulation board **31'** was positioned on rebar chairs **37** and centered left/right within the form; and the insulation board **31'** was inserted onto the horizontal rebar pieces **33** using predrilled holes **42** in the insulation board **31'** to secure the insulation board **31'** in the proper location;

(131) 7. In step (2) above, wythe tie pins **41** were installed in a grid pattern through pre-drilled holes **42** extending through the insulation board **31'**;

(132) 8. The pre-tensioned $\frac{3}{8}$ in cables **50** were installed thru (a) a pre-stressing cable anchor **51**, a first end support beam **52**, a first form end panel **23**, a second form end panel **24**, and out thru an

opposite second end support beam **53** along the back panel **23**, prior to installing the fence material **36**, the insulation **31**, and the rebar **32/33**, and (b) then looped back thru the opposite second end support beam **53**, the second form end panel **24**, the first form end panel **23**, the first end support beam **52**, and another pre-stressing cable anchor **51**, after installing the fence material **36**, the insulation **31**, and the rebar **32/33**;

(133) 9. The pre-stressing cable anchors **51** were fixed onto the free end abut to the first end support beam **52**;

(134) 10. A spacer (not shown) was positioned between the loop **50'** of the pre-stressing cable **50** and the second end support beam **53**;

(135) 11. The pre-stressing cables **50** were pulled “hand-tight” against the spacer so as to set the pre-stressing cable anchors **51** on both ends;

(136) 12. L-angle embeds **80** were attached to magnets (not shown) at top and bottom corners of the second form end panel **24** (i.e., the corners representing the lower corners of the finished pre-stressed insulated concrete panel **10**);

(137) 13. Plastic tubes **38** were inserted thru holes **40** along and proximate top and bottom edges **222/223** of the form front and back side panels **21/22** so as to protrude from front and back form side panels **21/22** (i.e., to form the holes **65** in the finished pre-stressed insulated concrete panel **10** used for mounting pour-in-place forms thereto);

(138) 14. A hydraulic tensioning cylinder (not shown) was installed so that the base was against the pre-stressing cable anchors **51** (i.e., along second support beam **53**) and the cylinder rod was nested within the loop **50'** of the pre-stressing cable **50** (Note that the tension cable **50** is typically routed so that each row or layer of tension cables **50** is actually one cable **50**. For example, the cable **50** could be installed starting on the right end of the form and could be inserted through the appropriate holes to the left end of the form where it would make a loop **50'** and then be inserted back into the left end of the form. Two cable anchors **51** would be installed in the correct orientation and then the cable **50** would be inserted through the appropriate holes back to the right side of the form. Cable anchors **51** would be installed on the ends of the tension cables **50** on the right side of the form. A tensioning device (not shown) would be installed between the loop **50'** in the cable **50** and the two cable anchors **51** on the left end of the form. As pressure is applied to the tension device it is positioned inside the loop **50'** of the tension cable **50** and pushes against the cable anchors **51** on the left end of the form. The cable anchors **51** on the right end of the form prevent the cable **50** from moving while pressure is applied on the other end of the cable **50** with the tensioning device. As the looped end of the cable **50'** is put under tension it stretches the two lengths of tension cable **50** in the form in unison. Once the correct tension is achieved the cable anchors **51** on the left end of the form (behind the tension device) holds the tensioned cable **50** under tension while the tension on the tension device is released.);

(139) 15. The cylinder was pressurized to the recommended pressure to achieve the desired pre-stressing cable tension;

(140) 16. The hydraulic tensioning cylinder was depressurized and moved to the next pre-stressing cable loop **50'**;

(141) 17. Steps (14) to (16) were repeated to tension all prestressing cables **50**;

(142) 18. Concrete **60** was poured into the form;

(143) 19. Vibrators (not shown) were energized to ensure concrete **60** flowed to fill entire form without leaving and voids;

(144) 20. Concrete **60** was allowed to cure (i.e., set) to a predetermined hardness prior to form disassembly;

(145) 21. After initial concrete **60** cure time had passed, pre-tension cables **50** were cut in the gaps between the form end panels **24/23** and the end structural beams **52/53** (i.e., along the bottom edge **61** and top edge **62** of the formed pre-stressed insulated concrete panel **10**);

(146) 22. The plastic tubes **38** were removed from along top and bottom edges **222/223** of the form

front and back side panels **21/22**;

(147) 23. Remove top connecting straps;

(148) 24. A specially designed lifting fixture (not shown) was installed to the rebar **33** at both ends of the side **66** of the formed pre-stressed insulated concrete panel **10**;

(149) 25. The lifting fixtures were connected to the boom of yard fork truck;

(150) 26. The front form side panel **21** was unbolted and removed;

(151) 27. The first and second form end panels **23/24** were unbolted and removed;

(152) 28. The formed pre-stressed insulated concrete panel **10** was lifted to (i) remove from the form, and (ii) transport to a curing rack **75**;

(153) 29. The lifting fixture was removed from the formed pre-stressed insulated concrete panel **10**; and

(154) 30. The formed pre-stressed insulated concrete panel **10** was allowed to fully cure prior to installation.

Example 2—Method for Forming a Building Using Pre-Stressed Insulated Concrete Panels

(155) The following method was used to form a building construction using a combination of (1) pre-stressed insulated concrete panels, such as the pre-stressed insulated concrete panels formed in Example 1, and (2) poured-in-place panels to form a continuous structural perimeter around an edge of a poured concrete floor and foundation. Although additional steps may be used, the steps used in the present examples were: 1. Pre-stressed insulated concrete panels were erected on a panel ledge on a perimeter of a previously poured concrete slab; 2. Pre-stressed insulated concrete panels were erected leaving a 1.0 foot (ft) gap between adjacent wall panels and supported by temporary bracing; 3. Pre-stressed insulated concrete panels were separated from one another with larger/wider gaps at locations intended for doors, sliding glass doors, or windows and supported by temporary bracing; 4. Pre-stressed insulated concrete panels included horizontal rebar that extended on either side of the pre-stressed panel; 5. Additional rebar was fastened to (i) the rebar extending from the sides of the pre-stressed insulated concrete panels in the 1.0 ft gaps between the wall panels, (ii) below and above the elevation of the window in the window gaps, and (iii) above the door and sliding glass door gaps; 6. Temporary pour-in-place forms were assembled (i) on the inside and outside of the 1.0 ft wall gaps, (ii) below & above the window gaps, and (iii) above the door and sliding glass door gaps; 7. The pour-in-place forms have features to mount electrical boxes for outlets and switches and conduit chases that extend to a top of a given gap, conduit chases consisting of rectangular boxes were installed in the given gap to provide a pathway for electrical wires for outlets and switches and provide dead air space insulation within the concrete of the gap; 8. Concrete was poured into (i) the forms in the 1.0 ft wall gaps, (ii) the forms below and above the window elevation in the window gaps, and (iii) the header forms above the door and sliding glass door gaps; 9. After the concrete in the gaps was cured, the forms and temporary bracing were removed and the continuous concrete perimeter wall remained; and 10. A steel roof was attached to the continuous concrete perimeter wall so as to provide a building wherein the concrete wall structure, the concrete slab, and the steel roof provided a building structure constructed from non-combustible and termite proof materials.

Example 3—Method for Forming a Building Using Pre-Stressed Insulated Concrete Panels

(156) The following method was also used to form a building construction using a combination of (1) pre-stressed insulated concrete panels, such as the pre-stressed insulated concrete panels formed in Example 1, and (2) poured-in-place panels to form a continuous structural perimeter around an edge of a poured concrete floor and foundation. Although additional steps may be used, the steps used in the present examples were: 1. Prepare the site; 2. Clear the land and grade; 3. Dig and form footer for garage and house slabs with panel ledge around perimeter and rough in the utilities; 4. Pour concrete footer and slab for garage and house; 5. Allow concrete footers and slabs to cure; 6. Layout panel location on panel ledge around perimeter; 7. Transport panel forms to site and stage on site slabs; 8. Stage concrete panel curing rack on site; 9. Form pre-stressed insulated concrete

panels using the method as described in Example 1; 10. After first batch of concrete panels are cured, re-attach lifting fixture or attach clamping device; 11. Rig lifting fixtures from both sides of the top of the panel to the boom on the yard fork truck, lift the panel with the boom of the fork truck as the rotary device turns the panel in the upright position thus transferring the load to the fork truck; 12. Transport the panel to its final position on the foundation panel ledge; 13. Level and plumb the panel and attach temporary bracing to the rebar extending from either side of the panel (Note that an extra tube may be added on each side at a distance from the edge that will be beyond the extent the pour-in-place forms overlap the edge and at an elevation suited to attach the temporary rigging. A shoulder eyebolt could be inserted thru the hole in the concrete and secured with a nut on the backside to which the temporary bracing is fastened.); 14. Anchor the panel to the foundation panel ledge; 15. Remove all rigging from the panel; 16. Repeat steps 10 to 15 for remaining panels; 17. After concrete panels are set and secured with temporary bracing, drill required holes into the panel ledge to epoxy vertical rebar pieces, as needed, in all pour-in-place 1.0 ft gaps; 18. Fasten horizontal rebar pieces (i.e., on the exterior side of the panels) to the horizontal rebar extending from the sides of the concrete panels into the gaps in the walls and the bottom and top of the window gaps, and the top of the door and sliding glass door gaps; 19. Install the electrical chase/insulation between the rebar in the 1.0 ft gaps between wall panels; 20. Fasten horizontal rebar pieces (i.e., on the interior side of the panels) to the horizontal rebar extending from the sides of the concrete panels into the 1.0 ft gaps in the walls; 21. Install and epoxy in place all required vertical rebars in the poured-in-place gaps; 22. Wire tie the vertical rebar pieces to the horizontal rebar pieces on the interior and exterior of the poured-in-place gaps; 23. Position electrical boxes and fasten to the gaps inside form, as applicable; 24. Set and secure in place the inside and outside gap forms for the 1.0 ft gaps and fasten to the wall by running bolts through the holes left by the tubes along the sides of the concrete panels formed in Example 1; 25. Install the header form above the windows, doors, and sliding glass door gaps; 26. Install temporary support chairs on the panel ledge below the windows, and on the header forms above the door and sliding glass door gaps; 27. Set insulation (e.g., an insulation board or other type of insulation) between the rebar on the temporary chairs below the windows and above the doors and sliding glass doors gaps; 28. Install horizontal rebar pieces through drilled holes in the insulation and secure to the horizontal rebar pieces on the inside and outside of the insulation, as applicable; 29. Remove temporary insulation support chairs, as applicable; 30. Install vertical rebar pieces, as required, on the inside and outside of the poured-in-place gaps above and below the windows, and above all the doors, and wire tie to all horizontal rebar pieces; 31. Install wythe anchor pins in grid pattern holes in insulation, as applicable; 32. Set and secure in place the inside and outside gap forms for all window and door gaps, and fasten to the wall by running bolts through the holes left by the plastic tubes along the sides of the form panels; 33. Install additional tie supports, as required, on poured-in-place gaps (e.g., window and doors) wider than 1 ft depending on finish sizes; 34. Pour concrete into the forms making sure to thoroughly vibrate the concrete to fill all the space and not leave any voids; 35. Allow concrete to cure before removing the forms; 36. Cut the rebar flush to the concrete (since it will not be used) where the windows, doors and sliding glass doors will be installed (i.e., where there would be no poured-in-place concrete); 37. Remove the temporary bracing; 38. Install and anchor steel components all around the perimeter of the top of the wall; 39. Rig and hoist steel roof trusses, and set into position; 40. Install temporary bracing, and weld the steel roof trusses to the anchor steel components; 41. After all roof trusses are installed, install steel roof purlins between the trusses; 42. Install cement decking prior to roofing installation; 43. Install vapor barrier prior to roofing installation; and 44. Install traditional standing seam metal roofing and trim.

Example 4—Method for Forming a Wall of a Building Using Pre-Stressed Insulated Concrete Panels

(157) The following method was also used to form a wall of a building structure (e.g., a single-family home) using a combination of (1) nine pre-stressed insulated concrete panels, such as the

pre-stressed insulated concrete panels formed in Example 1, the nine pre-stressed insulated concrete panels having a variety of panel widths, and (2) poured-in-place panels to form a continuous wall. Although additional steps may be used, the steps used in the present examples were:

(158) A. Clear the lot for the foundation. Since crane service is not needed, lot clearing can be minimized.

(159) B. Excavate for foundation footers and slab. Footers for the foundation were somewhat larger than a normal timber-built house.

(160) C. A monolithic pouring technique was used to pour the footers and the slab. A panel ledge slightly greater in width than wall panel thickness was incorporated into the footer.

(161) D. Once the footer and the slab had cured, the pre-stressed insulated concrete forms were placed. Forms and racks for storing panels were positioned to easily accommodate pouring the forms and storing the panels after initial curing for final curing.

(162) E. A string line was established at a given distance from the panel ledge. This step ensured that the wall was straight regardless of variations in the straightness or squareness of the footers/slab.

(163) F. Calculation were made to determine where the window opening was to be. A panel was set on one side of the window opening assuring that the side (edge) of the panel, which faces the window, was plum. At this point a hole was drilled in the panel ledge so that a piece of angle could be installed using an expanding anchor bolt. Once the angle was secured to the panel ledge, it was welded to the adjacent angle, which was embedded in the bottom corner of the panel.

Simultaneously, the panel was plumbed to the inside of the house. This was accomplished by attaching temporary bracing to the upper porting of the panel. Installing the angle and anchor bolt on the opposite edge of the panel completed the panel installation.

(164) G. At this point the other side of the window opening was installed. Panel installation continued using the same methodology outline in step F. Care was taken to ensure that the edge of the panel facing the window opening was plum. This ensured that the window opening was the correct dimension and square. The window opening extended from the finish floor to the top of the wall allowing great flexibility for window sizing.

(165) H. The balance of the panels on this wall were installed in a similar manner allowing approximately 12 inches between the panels. At this point, nine panels have been installed on the single wall with one window opening. There are two corner gaps of approximately 12 inches and the window opening. The following steps detail (1) turning a gap into an insulated column, (2) turning a gap into an insulated column with an electrical chase installed, and (3) addressing a gap within the footer and header above and below the window.

(166) I. Installing a column, which is insulated, but has no electrical chase:

(167) Each panel adjacent to what will become the column had form board holes approximately 2 inches from the column edge at appropriate intervals. This allowed the form boards to be bolted to the panel edge with no wall ties through the form boards (see, holes in FIG. 3A). Before bolting the form boards to the panel, the horizontal outside rebar was tied together. Once the outside rebar was tied together, an insulated chase was installed. The insulated chase was approximately 6 inches wide and 3 inches deep. It was wired to the horizontal rebar and stopped 16 inches from the bottom and 10 inches from the top. This allowed for 10 inches of solid concrete at the top and 16 inches at the bottom. Once the chase was wired in place the inside horizontal rebar was wired together. A minimum of two vertical rebars (e.g., four rebar) were then epoxied 3 inches deep in the panel ledge. Once the horizontal rebars were tied to the chase and the vertical rebars were epoxied in the panel ledge, the form boards were installed using the pre-engineered holes. Column was ready to be poured.

(168) J. Installing a column, which is insulated, and has an electrical chase:

(169) There are two major differences when an insulated chase is an electrical conduit. The first

difference is, an electrical conduit will be fastened to the end of the chase for the first 14 inches at the top of the column, 10 inches in the wall, 4 inches above the wall. Depending on the wire size to be pulled through the chase $\frac{1}{2}$ ", $\frac{3}{4}$ " or 1-inch electrical conduit (conduit non-metallic) can be used. The second major difference is an electrical box will be attached to the form boards before they are bolted to the panel. The electrical box will be aligned so that it abuts to the bottom of the insulated chase. This will allow 16 inches of solid concrete at the bottom of the column. Form boards for the columns can either be wood or aluminum. In either case, placement of the electrical box on the form boards is accurately determined.

(170) K. The headers over the windows used the same technique as the headers over the doors. The footer under the windows was similar to the headers, and resembled standard concrete form application. With the footers, rebar was needed to be tied and form boards installed. At the end of the forms, the form boards were attached through the pre-engineered holes. Depending on the size of the forms, additional wall ties might be required. Fiberglass wall ties were used to make sure no rust stains appeared. The same technique was used for the headers over windows and doors. In the case of headers, temporary support columns might be required. Fiberglass wall ties were used.

(171) L. Once all the columns, footers under the windows, and headers over the doors and windows were poured, the concrete structure was complete. At this point, reinforcing steel was added at the top of the panels. For the majority of the panels, flat bar steel of the correct width and thickness was used. At the gable ends of the house, flat bar steel was replaced with the appropriate size steel channel. Other steel including steel beams might be required depending on the size and shape of the house. The steel reinforcement was attached to the top of the panels by drilling the correct size hole and installing expansion bolts through the steel, which was used to fasten the steel to the concrete. The structure (house) was now complete less the roof and any required interior walls. (The wall structure could stay in this state indefinitely.)

(172) M. Installation of a fireproof roof utilized the following steps. m.1. Installation of metal trusses. m.2. Installation of metal purlins. m.3. Installation of fireproof roof panels. m.4. Installation of metal shingles or a standing seam roof. m.5. Installation of gable ends including fireproof panels and metal shingles or metal panels.

(173) N. To complete the project, interior walls, insulation for the roof, interior and exterior doors, windows, and finishing were installed.

(174) In addition, it should be understood that although the above-described methods of making pre-stressed insulated concrete panels, pre-stressed insulated concrete panels, and methods of using pre-stressed insulated concrete panels are described as "comprising" one or more components or steps, the above-described methods of making pre-stressed insulated concrete panels, pre-stressed insulated concrete panels, and methods of using pre-stressed insulated concrete panels may "comprise," "consists of," or "consist essentially of" the above-described method steps and components of the methods of making pre-stressed insulated concrete panels, the pre-stressed insulated concrete panels, and the methods of using pre-stressed insulated concrete panels. Consequently, where the present invention, or a portion thereof, has been described with an open-ended term such as "comprising," it should be readily understood that (unless otherwise stated) the description of the present invention, or the portion thereof, should also be interpreted to describe the present invention, or a portion thereof, using the terms "consisting essentially of" or "consisting of" or variations thereof as discussed below.

(175) As used herein, the terms "comprises," "comprising," "includes," "including," "has," "having," "contains," "containing," "characterized by," or any other variation thereof, are intended to encompass a non-exclusive inclusion, subject to any limitation explicitly indicated otherwise, of the recited components. For example, a method of making pre-stressed insulated concrete panels, a pre-stressed insulated concrete panel, and/or a method of using pre-stressed insulated concrete panels that "comprises" a list of elements (e.g., components or steps) is not necessarily limited to only those elements (or components or steps), but may include other elements (or components or

steps) not expressly listed or inherent to the method of making the pre-stressed insulated concrete panel, the pre-stressed insulated concrete panel, and/or the method of using the pre-stressed insulated concrete panel.

(176) As used herein, the transitional phrases “consists of” and “consisting of” exclude any element, step, or component not specified. For example, “consists of” or “consisting of” used in a claim would limit the claim to the components, materials or method steps specifically recited in the claim except for impurities ordinarily associated therewith (i.e., impurities within a given component). When the phrase “consists of” or “consisting of” appears in a clause of the body of a claim, rather than immediately following the preamble, the phrase “consists of” or “consisting of” limits only the elements (or components or steps) set forth in that clause; other elements (or components) are not excluded from the claim as a whole.

(177) As used herein, the transitional phrases “consists essentially of” and “consisting essentially of” are used to define a method of making a pre-stressed insulated concrete panel, a pre-stressed insulated concrete panel, and/or a method of using a pre-stressed insulated concrete panel that includes materials, steps, features, components, or elements, in addition to those literally disclosed, provided that these additional materials, steps, features, components, or elements do not materially affect the basic and novel characteristic(s) of the claimed invention. The term “consisting essentially of” occupies a middle ground between “comprising” and “consisting of.”

(178) Further, it should be understood that the herein-described methods of making pre-stressed insulated concrete panels, pre-stressed insulated concrete panels, and/or methods of using pre-stressed insulated concrete panels may comprise, consist essentially of, or consist of any of the herein-described components, method steps, and/or features, as shown in the figures with or without any feature(s) not shown in the figures. In other words, in some embodiments, the methods of making pre-stressed insulated concrete panels, the pre-stressed insulated concrete panels, and/or the methods of using pre-stressed insulated concrete panels of the present invention do not have any additional features other than those shown in the figures, and such additional features, not shown in the figures, are specifically excluded from the methods of making pre-stressed insulated concrete panels, the pre-stressed insulated concrete panels, and/or the methods of using the pre-stressed insulated concrete panels. In other embodiments, the methods of making pre-stressed insulated concrete panels, the pre-stressed insulated concrete panels, and/or the methods of using the pre-stressed insulated concrete panels of the present invention do have one or more additional features that are not shown in the figures.

(179) The present invention is described above and further illustrated below by way of claims, which are not to be construed in any way as imposing limitations upon the scope of the invention. On the contrary, it is to be clearly understood that resort may be had to various other embodiments, modifications, and equivalents thereof which, after reading the description herein, may suggest themselves to those skilled in the art without departing from the spirit of the present invention and/or the scope of the appended claims.

Claims

1. A method of making a wall structure of a building, said method comprising: forming a pre-stressed insulated concrete panel with a concrete panel thickness of the pre-stressed insulated concrete panel extending in a horizontal orientation, a concrete panel width of the pre-stressed insulated concrete panel extending in a vertical orientation, and a concrete panel length of the pre-stressed insulated concrete panel extending in a horizontal orientation, said forming step taking place on a base form at a job site of the building; and moving the pre-stressed insulated concrete panel from the base form on the job site to a wall location of the building on the job site, wherein said forming step comprises: forming the pre-stressed insulated concrete panel on (i) the base form and between (ii) front and back side panels extending along the concrete panel width and the

concrete panel length, and (ii) first and second form end panels extending along the concrete panel width and the concrete panel thickness, and wherein the base form comprises (a) one or more first base form holes within an upper surface of the base form, and said forming step comprises attaching the first and second form end panels to an upper surface of the base form via the one or more first base form holes, and (b) one or more second base form holes within an upper surface of the base form, and said forming step comprises attaching the front and back side panels to the base form via the one or more second base form holes.

2. The method of claim 1, wherein (A) each of the front and back side panels has (a) a side panel height of from about 30 inches (in) to about 72 in, and (b) a side panel length of from about 8 ft to about 12 ft, and (B) each of the first and second form end panels (a) has an end panel height of from about 30 in to about 72 in, and (b) an end panel width of from about 6 in to about 16 in.

3. The method of claim 1, wherein the base form has (a) a base form length of from about 10 feet (ft) to about 24 ft, and (b) a base form width of from about 12.0 in to about 24 in.

4. The method of claim 1, wherein said forming step further comprises: positioning one or more embedded panel components within a form volume bound by (i) an upper surface of the base form, (ii) inner surfaces of the front and back side panels, and (iii) inner surfaces of the first and second form end panels, wherein the one or more embedded panel components comprise (i) an insulation board, (ii) vertical rebar pieces, (iii) horizontal rebar pieces, (iv) vertical rebar pieces and horizontal rebar pieces arranged in at least one layer of rebar, (v) at least one layer of fence material, (vi) from about 3 to about 8 chair supports, (vii) from about 6 to about 16 pieces of pipe material positioned thru holes along and proximate top and bottom edges of the front and back side panels, (viii) a plurality of wythe tie pins in a grid pattern through pre-drilled holes extending through the insulation board, (ix) a plurality of pre-stressing members, or (x) any combination of (i) to (ix).

5. The method of claim 4, wherein the one or more embedded panel components comprise the insulation board sandwiched between two layers of rebar, and the two layers of rebar are sandwiched between two layers of fence material.

6. The method of claim 4, wherein the plurality of pre-stressing members comprises a plurality of pre-stressing cables.

7. The method of claim 1, wherein said forming step further comprises: positioning a plurality of pre-stressing members so as to extend thru (a) a first pre-stressing member anchor, a first end support beam, the first form end panel, the second form end panel, and out thru an opposite second end support beam; applying tension on the plurality of pre-stressing members; pouring concrete into a form comprising (i) the base form, (ii) front and back side panels extending along the concrete panel width and the concrete panel length, and (iii) first and second form end panels extending along the concrete panel width and the concrete panel thickness.

8. A wall structure formed in the method of claim 1.

9. A wall structure formed in a method comprising: forming a pre-stressed insulated concrete panel with a concrete panel thickness of the pre-stressed insulated concrete panel extending in a horizontal orientation, a concrete panel width of the pre-stressed insulated concrete panel extending in a vertical orientation, and a concrete panel length of the pre-stressed insulated concrete panel extending in a horizontal orientation, said forming step taking place on a base form at a job site of the building; and moving the pre-stressed insulated concrete panel from the base form on the job site to a wall location of the building on the job site, wherein: the pre-stressed insulated concrete panel comprises one or more embedded panel components, said one or more embedded panel components comprising one or more of: (i) an insulation board, (ii) vertical rebar pieces, (iii) horizontal rebar pieces, (iv) vertical rebar pieces and horizontal rebar pieces arranged in at least one layer of rebar, (v) at least one layer of fence material, (vi) from about 3 to about 8 chair supports, (vii) from about 6 to about 16 pieces of pipe material positioned thru holes along and proximate top and bottom edges of the front and back side panels, (viii) a plurality of wythe tie pins in a grid pattern through pre-drilled holes extending through the insulation board, (ix) a plurality of pre-

stressing members, or (x) any combination of (i) to (ix).

10. The wall structure of claim 9, wherein the one or more embedded panel components comprise (1) an insulation board, (2) vertical rebar pieces and horizontal rebar pieces arranged in at least one layer of rebar, and (3) at least one layer of fence material, wherein the insulation board is sandwiched between two layers of rebar, and the two layers of rebar are sandwiched between two layers of fence material, and the two layers of fence material are sandwiched between two layers of pre-stressing members.

11. The wall structure of claim 10, wherein the concrete panel width ranges from about 24 in to about 72 in, the concrete panel length ranges from about 72 in to about 130 in, and the concrete panel thickness ranges from about 5.0 in to about 14.0 in.

12. A building comprising the wall structure formed in the method of claim 1, wherein the building comprises a residential building, a garage, a barn, a shed, a utility building, an electrical/computer housing building, a commercial building, an industrial building, any other structure to be insulated, etc.

13. The building of claim 12, wherein the building comprises a residential building.

14. The wall structure of claim 9, wherein the plurality of pre-stressing members comprises a plurality of pre-stressing cables.

15. A method of making a wall structure of a building, said method comprising: forming a pre-stressed insulated concrete panel having a concrete panel thickness, a concrete panel width, and a concrete panel length, the pre-stressed insulated concrete panel comprising concrete and a plurality of tensioned prestressing members extending through the concrete, said forming step taking place on a base form at a job site of the building; and moving the pre-stressed insulated concrete panel from the base form on the job site to a wall location of the building on the job site, wherein: the concrete panel thickness extends in a horizontal orientation, the concrete panel width extends in a vertical orientation, and the concrete panel length extends in a horizontal orientation, and said plurality of tensioned prestressing members extend along the concrete panel length in a horizontal orientation during said forming step; the forming step comprises forming the pre-stressed insulated concrete panel on (i) the base form and between (ii) front and back side panels extending along the concrete panel width and the concrete panel length, and (iii) first and second form end panels extending along the concrete panel width and the concrete panel thickness, and the base form comprises (a) one or more first base form holes within an upper surface of the base form, and said forming step comprises attaching the first and second form end panels to an upper surface of the base form via the one or more first base form holes, and (b) one or more second base form holes within an upper surface of the base form, and said forming step comprises attaching the front and back side panels to the base form via the one or more second base form holes.

16. The method of claim 15, wherein said forming step further comprises: positioning a plurality of pre-stressing members so as to extend thru (a) a first pre-stressing member anchor, a first end support beam, the first form end panel, the second form end panel, and out thru an opposite second end support beam; applying tension on the plurality of pre-stressing members; pouring concrete into a form comprising (i) the base form, (ii) front and back side panels extending along the concrete panel width and the concrete panel length, and (iii) first and second form end panels extending along the concrete panel width and the concrete panel thickness.

17. A wall structure of a building formed in the method of claim 15.

18. The wall structure of claim 17, wherein the building comprises a residential building, a garage, a barn, a shed, a utility building, an electrical/computer housing building, a commercial building, an industrial building, any other structure to be insulated, etc.

19. The method of claim 15, wherein the plurality of pre-stressing members comprises a plurality of pre-stressing cables.

20. The method of claim 7, wherein the plurality of pre-stressing members comprises a plurality of pre-stressing cables.

